COLDCHAIN

COMPETITION / FIRST ROUND

Preserve every degree.

Vaccine & medicine storage monitor: two-fridge IoT control, live intelligence, traceable inventory, and tamper-evident security.

100-POINT TASK COVERAGE

Technical implementation dossier

Delivered: running FastAPI + responsive web application; ESP32 two-node firmware source; MQTT message signing and retry; budget logic; inventory/QR; a trained synthetic forecast model; automated tests; technical documentation.

FRIDGE NODES	AUTOMATED TESTS	FORECAST HORIZON
02	13 PASS	30 MIN

Prototype only • Synthetic ML evaluation • Hardware validation required

01 / DESIGN

One connected safety system

Separate edge safety, signed device transport, operational records, and human decisions. Both fridges are independent physical targets.

01

EDGE / ESP32 + DS18B20

Two identical firmware nodes with unique device IDs and secrets, OLED status, LED/buzzer, door switch, relay, battery and 5-second sampling.

02

SIGNED TRANSPORT / MQTT

HMAC-SHA256, timestamp, random nonce, sequence deduplication, flash queue, authenticated application ACK, and retained red-latch command.

03

SAFETY ENGINE / FASTAPI

Atomic ingestion, 60-minute weighted exposure for every batch, door alerts, restricted admin discard reset and relocation validation.

04

INTELLIGENCE / MACHINE LEARNING

30-minute RandomForest temperature projection and compressor cooling-rate health score; simulations are clearly labeled.

05

WORKSPACE / WEB + PUBLIC QR

Role-based live dashboard; pharmacist move workflows; inspector read-only access; public lot/expiry/status via opaque QR URL.

DESIGN BOUNDARY

Local relay timing and local alarm do not depend on the cloud. Machine-learning suggestions cannot operate the physical compressor or independently authorize the movement of real medicine.

02 / SAFETY LOGIC

Safety is deterministic

Relay decisions remain local. Heat exposure is accounted for per batch and its history is preserved when inventory moves.

RELAY SWITCH ON	RELAY SWITCH OFF	DOOR ALARM
≥ 7°C	≤ 3°C	> 60 s
Only if at least 60 s since switch-off	Prevents over-cooling; includes min-off safety	Buzzer plus configured external alert

HEAT-DOSE MODEL / SINGLE INTERVAL

T ≤ 8°C	8 < T ≤ 12°C	T > 12°C
+0 seconds	+1 × elapsed	+2 × elapsed

Every valid interval is weighted using the preceding measured temperature; long data gaps are flagged, never silently assumed safe.

BATCH DECISION RULES

SAFE	20% or more budget remaining
AT RISK	Less than 20% budget remaining
DISCARD	0% remaining — software latch persists until restricted admin reset

CLINICAL SAFETY NOTE

The reset action does not restore vaccine potency. Refrigeration requires validated temperature sensors, certified external compressor controls, manufacturer-specific stability procedures, and qualified pharmacist oversight. The source code was not tested with live mains equipment.

03 / TRUST & WORKFLOWS

Evidence across every action

Inventory events, sensor measurements and security outcomes are linked in a tamper-evident audit history.

01
DEVICE TRUST
HMAC-SHA256 signed packets; fresh send timestamp, random nonce, per-fridge sequence and changed-payload detection.

02
STAFF ACCESS
Six-hour JWT and bcrypt credential hashes. Admin, Pharmacist, Inspector permissions are enforced by server-side checks.

03
TAMPER DETECTION
SHA-256 hash chain over audit events; Verify Integrity also checks sensor rows against matching hash-covered records.

04
PUBLIC QR
Opaque URL token reveals only lot, product name, status, expiry and remaining exposure — no patient identifiers.

05
ALERT DELIVERY
Door >60 seconds generates alert attempt. Outbox records delivered / failed / not_configured accurately.

06
RELOCATION
One-click authorized confirmation after selecting a fresh, powered, closed and sufficiently cool destination.

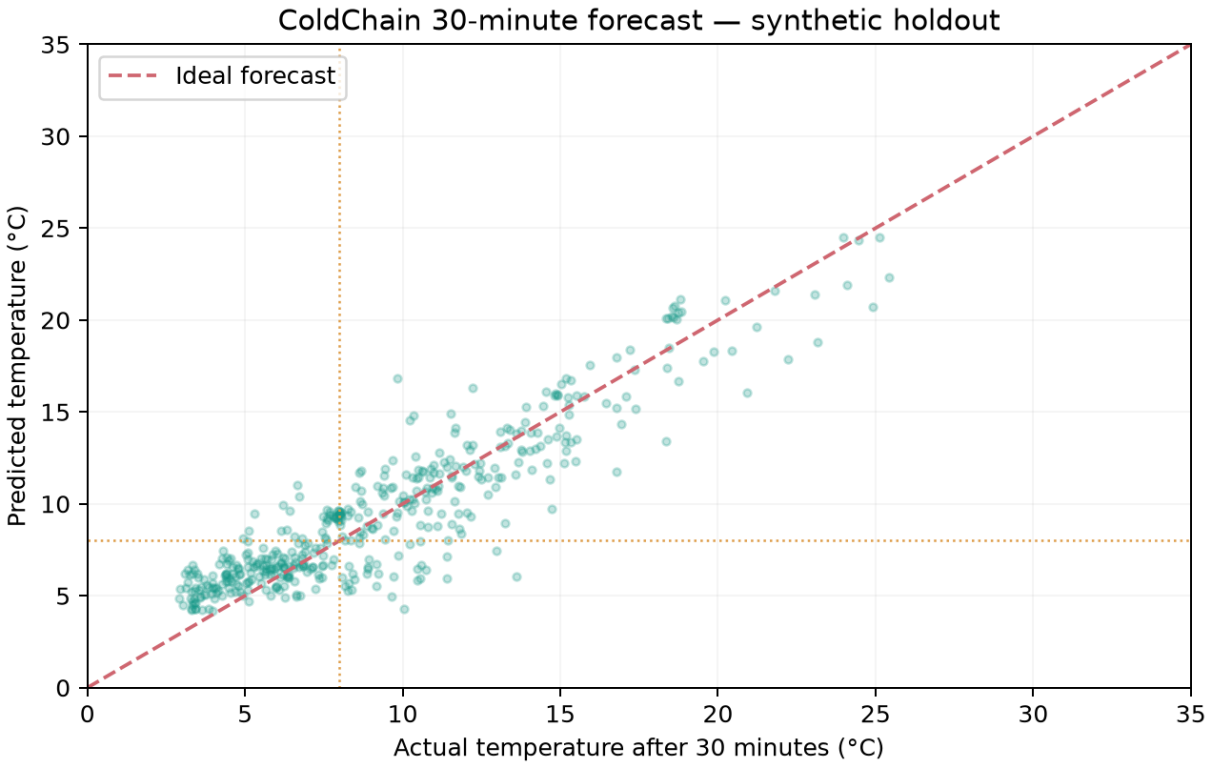
Privacy-first public QR • Read-only Inspector • Downloadable CSV evidence

04 / PREDICTIVE INTELLIGENCE

30 minutes ahead

Train/test split by complete simulated refrigeration trajectory to avoid overlap leakage between nearby windows.

TRAINING WINDOWS	HELD-OUT WINDOWS	MAE	BREACH RECALL
9,405	2,695	1.478°C	87.0%



RESULTS / SYNTHETIC TEST ONLY

220 independently seeded trajectories; 171 for training and 49 for testing. RandomForest, 30 trees (depth 10), eight input features (current temperature, trend, mean, standard deviation, door, mains, compressor state and compressor duty). Prediction breach classifier: precision 86.6%; recall 87.0%; F1 86.8%.

Important: The metrics are obtained solely from a reproducible synthetic simulator. No real medical fridge holdout or deployment testing was done. An engineering-rule fallback is labeled separately if the model is unavailable.

05 / DELIVERABLES

Tested and packaged

A runnable local demo and auditable source, with important remaining physical and external-service verification called out.

AUTOMATED TESTS	DEMO FRIDGES	FORECAST
13 / 13	2	1.93°C
Passed against isolated SQLite database and FastAPI test client	Live signed synthetic samples in HTTP smoke test	Synthetic holdout RMSE; no field validation

WINDOWS / QUICK START

```
py -3.11 -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python run_demo.py
```

LOCAL DEMO ACCOUNTS

Admin: admin / ColdChainDemo_Admin2026!

Pharmacist: pharmacist / ColdChainDemo_Pharmacist2026!

Inspector: inspector / ColdChainDemo_Inspector2026!

JUDGE VERIFICATION PENDING

- □ Flash and verify two physical ESP32s, OLED, relay thresholds, flash replay
- □ Calibrate temperature probes and battery sensing
- □ Configure Telegram/SMTP and verify real message delivery
- □ Review in a normal browser on desktop/tablet/mobile
- □ Validate forecast on independent measured refrigerators

Package index: `README.md`, `coldchain/`, `web/`, `firmware/`, `train_model.py`, `models/`, `tests/`, `docs/`, and `report/model_metrics.json`. The original round-1 PDF contains no task-specific point allocation and the author cannot guarantee a 100/100 judge score.